

Test Evidence: Integer Data Analyzer

Command

python3 homework_solution.py

Terminal Output

```
● aimaankhan@Aimaans-MacBook-Pro Lecture_1_export % /opt/homebrew/bin/python3 /Users/aimaankhan/SJ
olution.py
INTEGER DATA ANALYZER
-----
Enter integers separated by spaces: 10 20 11 12 124 42 20 42 11 11 22

----- Analysis Results -----
Count: 11
Minimum: 10
Maximum: 124
Sum: 325
Average: 29.545454545454547
Duplicates: [20, 11, 42]

Frequencies:
10: 1
20: 2
11: 3
12: 1
124: 1
42: 2
22: 1
❖ aimaankhan@Aimaans-MacBook-Pro Lecture_1_export %
```

Command

python3 homework_test_analyzer.py

Terminal Output

```
aimaankhan@Aimaans-MacBook-Pro Lecture_1_export % /opt/homebrew/bin/python3 test_analyzer.py
```

```
Test 1: Parsing well-formed input
```

```
Input: '10 20 30'
```

```
Output: [10, 20, 30]
```

```
PASSED
```

```
Test 2: Parsing negative numbers and extra whitespace
```

```
Input: '-10 20 -5 '
```

```
Output: [-10, 20, -5]
```

```
PASSED
```

```
Test 3: Parsing rejects non-integer input
```

```
Input: '10 abc 30'
```

```
Raised: ValueError: 'abc' is not a valid integer.
```

```
PASSED
```

```
Test 4: Parsing rejects empty input
```

```
Input: ''
```

```
Raised: ValueError: No integers were entered.
```

```
PASSED
```

```
Test 5: Analyzer computes core statistics
```

```
Input: [10, 20, 30, 20]
```

```
Count: 4
```

```
Minimum: 10
```

```
Maximum: 30
```

```
Sum: 80
```

```
Average: 20.0
```

```
Duplicates: [20]
```

```
PASSED
```

```
Test 6: Analyzer computes frequency map
```

```
Frequencies: {10: 1, 20: 2, 30: 1}
```

```
PASSED
```

```
Test 7: Analyzer handles negative values and duplicates
```

```
Input: [-10, 20, -5, 30, -10]
```

```
Count: 5
```

```
Minimum: -10
```

```
Maximum: 30
```

```
Sum: 25
```

```
Average: 5.0
```

```
Duplicates: [-10]
```

```
Frequencies: {-10: 2, 20: 1, -5: 1, 30: 1}
```

```
PASSED
```

```
Test 8: Analyzer rejects an empty list
```

```
Input: []
```

```
Raised: ValueError: Cannot analyze an empty list.
```

```
PASSED
```

```
=====  
All 8 tests passed!
```

```
❖ aimaankhan@Aimaans-MacBook-Pro Lecture_1_export %
```